Next phase plan: tasks, ranks, 10-seed costs and recalibration

Feasible Rank Window project · revised 25 October 2026 (v3: AG News and MNLI removed; rank grid 4–256) · based on the frw-v4 results (1,088 runs, docs/FINDINGS_V4.md)

Why change the tasks

On RTE, MNLI and SST-2, LoRA rank 1 already learned every task to within about 1 point of the best rank for all four models, so accuracy was flat across ranks. Classification tasks of this kind have a very low intrinsic dimension (Aghajanyan et al., ACL 2021). A rank window can only show real edges on tasks that need more adaptation capacity.

1. Task sequence (revised)

#	Task	Dataset	Capacity needed	Train / eval	Metric
1	Extractive question answering	SQuAD v1.1	medium	4,000 / 500	exact match / F1
2	Dialogue summarisation	SAMSum	high	4,000 / 500	ROUGE-L
3	Maths word problems (step by step)	GSM8K	high	4,000 / 500	exact match (final answer)

• **One format for every task:** the model generates the answer as text through its own output layer, so there are no per-task classifier heads (in v4 these caused most of the measured forgetting).

• **Equal data:** 4,000 training examples per task; a fixed 500-example evaluation subset per task.

• **General-knowledge check:** a fixed 500-question MMLU subset (answer-likelihood scoring) after each task.

• **Two task orders:** SQuAD → SAMSum → GSM8K, and the reverse.

Decision on AG News and MNLI: removed

Why removing them is valid (your position).

• The v4 data already answer the question for classification: r = 1 learns RTE, MNLI and SST-2 to within about 1 point of the best rank. AG News (4-way topic classification) is easier still. Rank cannot show a window on tasks that saturate at the smallest rank.

• Every run spends about 40% of its compute on these two tasks. Removing them cuts the cost per run and puts the whole budget on tasks where rank can matter.

• Since SQuAD, SAMSum and GSM8K all need real capacity, any rank effect on forgetting is not diluted by tasks that need almost nothing.

What is lost (my side), and how to handle it.

• **No within-study control.** If the window moves to higher ranks, part of the move could come from the new setup (text generation instead of heads, all-linear LoRA) rather than from task difficulty. A low-capacity task in the same sequence would separate the two. Mitigation: argue from v4 (same models, classification, flat curve) and state the change of setup as a limitation; or add MNLI back as a fourth task only at 3 ranks (4, 64, 1024) for about +15% cost.

• **A shorter sequence.** Three tasks gives two 'after' measurements for the first task, the same as v4. To study how forgetting builds up over many fine-tunings, a fourth high-capacity task would be better. A cheap option is WebNLG (data-to-text generation, short inputs), for about +30% cost.

• My recommendation: drop AG News without reservation. MNLI is optional; the plan below assumes it is dropped, as you decided.

2. Rank grid: 4, 8, 16, 32, 64, 128, 256

Why this grid is a good choice.

• Factor-2 spacing locates the edges of the window within a factor of 2, across seven ranks.

• It stays in the genuine low-rank regime. With all-linear LoRA, r = 256 trains 16–19% of the model's parameters; r = 512 and 1024 would have trained 29–77%, which is closer to partial full fine-tuning than to LoRA.

• Every rank fits a 48 GB L40S for all four models (adapter training state at r = 256 is 2.9–7.7 GB), so no 80 GB GPUs are needed and all runs use one GPU type, as in v4.

• It removes the two slowest and most expensive rank settings, which also showed the most instability in v4 and E1.

• Dropping r = 1 and 2 is reasonable: these generative tasks very likely need more than rank 2.

Remaining limitations. Trainable parameters of all-linear LoRA:

Model	r = 4	r = 16	r = 64	r = 256	Adapter training state at r = 256*
Llama-3.2-1B	2.8 M (0.2%)	11 M (0.9%)	45 M (3.6%)	180 M (15%)	2.9 GB
Qwen2.5-1.5B	4.6 M (0.3%)	18 M (1.2%)	74 M (4.8%)	295 M (19%)	4.7 GB
Granite-3.3-2B	7.0 M (0.3%)	28 M (1.1%)	113 M (4.5%)	451 M (18%)	7.2 GB
Qwen2.5-3B	7.5 M (0.2%)	30 M (1.0%)	120 M (3.9%)	479 M (16%)	7.7 GB

* fp32 adapter weights + gradients + two Adam moments (16 bytes per parameter); excludes the frozen model and activations. Percentages are of the total model size.

• **The upper edge may be open.** If accuracy is still rising at r = 256 for a task, the result can only say '256 or above'. Decision rule: if a model's best rank is 256, add r = 512 for that model only (about +15% of its cost) before writing conclusions.

• **The H-LoRA region is no longer tested.** Their 'two-thirds of hidden width' (≈1,024–1,365 here) is out of range, so the study cannot confirm or reject it directly. v4 already showed that region gives no benefit on classification.

• **Some matrices saturate at r = 256 in Qwen.** Qwen's k and v projections are only 256 wide, so at r = 256 they are already at full rank. Record the effective rank per matrix, as v4 did.

• **The lower edge is open.** If r = 4 is already inside the window, the result can only say '4 or below'.

3. Cost estimate with 10 seeds

Scaled from measured v4 run times. The 3 generative tasks total about 0.85× the training tokens of the earlier 5-task plan (4,000 examples × ~210 tokens each), plus about 4 minutes of text-generation evaluation. All-linear LoRA adds about 30%, and r = 256 about 12% more. All runs on L40S at $2.17/h including CPU and memory.

Model	Per run	Reported: 7 ranks × 10 seeds	LR search: 3 ranks × 4 LRs × 2 seeds	Total
Llama-3.2-1B	~$0.70	~$50	~$14	~$64
Qwen2.5-1.5B	~$1.05	~$75	~$20	~$95
Granite-3.3-2B	~$1.75	~$125	~$34	~$159
Qwen2.5-3B	~$1.75	~$125	~$34	~$159
All four models				**~$477**

Schedule against $180 per month	Work	Estimated cost
Month 1	Pilot (~$5), epoch check (~$15), Llama-3.2-1B	~$84
Month 2	Qwen2.5-1.5B	~$95
Month 3	Granite-3.3-2B	~$159

Schedule against $180 per month	Work	Estimated cost
Month 4	Qwen2.5-3B	~$159
Optional	r = 512 for a model whose best rank is 256 (+~15% of that model); MNLI control at 3 ranks (+~15%); second task order (+60–70%); WebNLG 4th task (+~30%)	—

Uncertainty: treat every figure as ±40%. Generation length is the least certain part; a ~$5 pilot on one model at r = 16 and r = 256 should come first. Months 1 and 2 together (~$179) could fit one month's $180, but with no safety margin, so the plan keeps them separate.

4. Hyperparameters that need recalibrating

Hyperparameter	Recalibrate?	Reason
Learning rate	Yes (required)	It was the strongest driver of forgetting in every experiment. The loss (text generation), tasks and target modules all change. Search 4 LRs × 2 seeds at r = 4, 32 and 256 (both ends of the grid and the middle): E1 showed high ranks can need about half the LR. Use the lowest learning rate within one standard error of the best, and interpolate for the ranks in between.
Epochs per task	Yes (quick check)	3 epochs was confirmed for classification only. One check at r = 16 comparing 2 / 3 / 4 epochs with 5 seeds, about $15.
Max sequence length	Yes	Raise from 256 to 512 tokens; SQuAD, SAMSum and GSM8K exceed 256.
Evaluation decoding	Set, not tuned	Greedy decoding; maximum new tokens per task: 32 for SQuAD, 96 for SAMSum, 256 for GSM8K.
Classifier head learning rate	No longer needed	The generative format has no classifier heads.
α = 32 (scale α/r), AdamW (0.9, 0.999, 1e-8), batch 16, warmup 6% + cosine, clipping 1.0, weight decay 0, dropout 0	No	Conventions or settings validated in v4; re-tuning them adds cost without clear benefit.
Target modules / seeds	No (decided)	All linear layers; 10 reported seeds, with 2 separate tuning seeds.

Sources

Measured costs and timings: results-v4/analysis (runs.csv, compute_by_model_and_stage.json). Findings: docs/FINDINGS_V4.md, sections 1–6 (E1: high ranks need a lower LR; E2: all-linear LoRA; E3: 3 epochs). Parameter counts are computed from each model's layer sizes. Intrinsic dimension: Aghajanyan, Zettlemoyer & Gupta (2021), ACL. Pricing: Modal list prices on 25 October 2026.